

System Complexity — Attendance FYP

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Feature	Complexity	Description	Reason for Complexity
Attendance Capture (QR + modes)	High	Generates/scans QR and manages live attendance sessions across online/physical modes. Prevents duplicates and ensures session integrity.	Concurrency, device variability, anti-replay protections, timing windows, error handling.
Geolocation Validation	Medium	Captures and validates location during attendance flows.	Permission handling, GPS accuracy variance, spoofing safeguards, fallback logic.
Fraud Detection & Alerts	High	Flags anomalies (e.g., suspicious patterns) and renders detection charts; supports alerting workflows.	Rule/threshold tuning, minimizing false positives, performant aggregation, alert lifecycles.
Data Validation & Integrity	High	Enforces uniqueness, status transitions, and manual overrides for attendance records.	Real-time checks, DB constraints, edge cases for time windows, consistency across clients.
Reporting & Export	Medium	Builds absence/fraud/attendance reports with filters and pagination.	Export formatting, dynamic query parameters, large result sets.
Analytics Dashboards	Medium	Trends, stats, and top absence reasons with interactive charts.	Real-time aggregations, rendering performance, chart library nuances.
Class & Schedule Management	Medium	Create/edit classes and schedules with validations.	Overlap detection, recurrence handling, complex UI state management.
Email Notifications	Medium	Sends post-lecture absence emails via Supabase function and client triggers.	Asynchronous execution, rate limits, retries, configuration management.

Authentication & Session Management	Medium	Login/signup/reset with Supabase; manages sessions securely.	Session persistence, error states, secure storage, edge-case authentication flows.
Absence Document Uploads (MC)	Medium	Uploads and validates medical certificates with metadata linking.	File type/size checks, secure storage, verification workflow integration.

Key Implementation References

- Attendance: `src/components/Event/QRGenerator.jsx`, `src/components/Event/AttendanceSession.jsx`, `src/pages/Dialogs/onlineAttendanceDialog.jsx`
- Location: `src/components/LocationInput.jsx`, `src/utills/geolocationUtils.js`
- Fraud: `src/components/Event/FraudDetectionChart.jsx`, `src/components/FlaggedAttendanceList.jsx`, `src/utills/fraudUtils.js`, `supabase/migrations/20251220_create_fraud_detection_settings_table.sql`, `supabase/migrations/20251220_create_fraud_detection_alerts_table.sql`
- Data Integrity: `src/utills/attendanceUtils.js`, `supabase/migrations/20251220_update_attendance_record_unique_constraint.sql`, `supabase/migrations/20251220_allow_flagged_status_in_attendance_record.sql`, `supabase/migrations/20251220_add_marked_manually_to_attendance_record.sql`
- Reporting/Analytics: `src/components/Event/ReportGenerator.jsx`, `src/pages/ReportsPage.jsx`, `src/components/FraudTable.jsx`, `src/components/AbsenceTable.jsx`, `src/pages/AnalyticsPage.jsx`, `src/components/AttendanceStats.jsx`, `src/components/Event/AttendanceTrends.jsx`, `src/components/Event/TopAbsenceReason.jsx`
- Classes/Scheduling: `src/pages/ManageClasses.jsx`, `src/components/Event/AddClassDialog.jsx`, `src/components/Event/EditClassDialog.jsx`, `src/components/ScheduleInput.jsx`
- Email: `supabase/functions/send-absence-email/index.ts`, `src/utills/sendAbsenceAfterLectureEnd.js`, `supabase/migrations/20251220_create_email_settings_table.sql`
- Auth: `src/pages/Login.jsx`, `src/pages/SignUp.jsx`, `src/pages/ForgotPassword.jsx`, `src/config/supabaseClient.js`
- Absence Docs: `src/pages/SubmitAbsenceDocumentPage.jsx`, `src/components/MCSubmission.jsx`

This document provides a one-page overview of functional areas and their complexity drivers to support planning, estimation, and risk assessment.

Proof Snippets (Code & DB References)

- Attendance Capture: [src/components/Event/AttendanceSession.jsx](#) – QR string includes class/session/password; expiry and end-time updates; fraud monitor start/cleanup.
- QR Generation: [src/components/Event/QRGenerator.jsx](#) – Encodes type/course/password; clipboard copy utility.
- Online Mode Payload: [src/pages/Dialogs/onlineAttendanceDialog.jsx](#) – Recording link + quiz JSON capture.
- Geolocation Utilities: [src/utills/geolocationUtils.js](#) – High-accuracy GPS with timeout/error codes.
- Geolocation UI: [src/components/LocationInput.jsx](#) – Debounced Nominatim search, GPS fetch, map preview.
- Fraud Detection Engine: [src/utills/fraudUtils.js](#) – Haversine distance, time-window anomaly detection, realtime supabase channel, alert upsert.

- Fraud Settings/Alerts Schema: [supabase/migrations/20251220_create_fraud_detection_settings_table.sql](#), [supabase/migrations/20251220_create_fraud_detection_alerts_table.sql](#).
- Fraud Visuals & Triage: [src/components/Event/FraudDetectionChart.jsx](#), [src/components/FraudTable.jsx](#), [src/components/FlaggedAttendanceList.jsx](#).
- Data Integrity Constraints: [supabase/migrations/20251220_update_attendance_record_unique_constraint.sql](#), [supabase/migrations/20251220_allow_flagged_status_in_attendance_record.sql](#), [supabase/migrations/20251220_add_marked_manually_to_attendance_record.sql](#).
- Attendance Metrics: [src/utils/attendanceUtils.js](#) – Rate calculations across sessions/enrollments.
- Reporting/Export: [src/components/Event/ReportGenerator.jsx](#) – Class/session/student CSV pivots.
- Analytics: [src/pages/AnalyticsPage.jsx](#), [src/components/Event/AttendanceTrends.jsx](#), [src/components/Event/TopAbsenceReason.jsx](#), [src/components/AttendanceStats.jsx](#).
- Classes & Scheduling: [src/pages/ManageClasses.jsx](#), [src/components/Event/AddClassDialog.jsx](#), [src/components/Event/EditClassDialog.jsx](#), [src/components/ScheduleInput.jsx](#).
- Email Notifications: [supabase/functions/send-absence-email/index.ts](#), [src/utils/sendAbsenceAfterLectureEnd.js](#), [supabase/migrations/20251220_create_email_settings_table.sql](#).
- Auth Flows: [src/pages/Login.jsx](#), [src/pages/SignUp.jsx](#), [src/pages/ForgotPassword.jsx](#), [src/config/supabaseClient.js](#).
- Absence Docs: [src/pages/SubmitAbsenceDocumentPage.jsx](#), [src/components/MCSubmission.jsx](#) – Upload, storage URL, review lifecycle.